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w/AMENDMENT 2
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SUPERSEDING
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MILITARY SPECIFICATION

TELESCOPE, PANORAMIC: M137, M137A1, M137A2 AND M137A3

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification covers four types of panoramic telescopes which provide four-power magnification and a ten-degree field of view. They are equipped with numerical counters which indicate azimuth settings and corrections. A settable counter provides readings from deflection reference point (M137, M137A1, M137A2 and M137A3) and a bearing reference point. The reticle pattern and counters numerals are illuminated.

1.2 Classification. The Panoramic Telescope is primarily intended to lay the howitzer in azimuth for indirect fire and there are the following types:

1.2.1 M137. This Panoramic Telescope is intended to support the M198 Howitzer and uses tritium.

1.2.2 M137A1. This Panoramic Telescope is intended to support the M119A1 Howitzer and uses tritium.

1.2.3 M137A2. This Panoramic Telescope is intended to support the M119 and M777 Series Howitzer and does not use tritium.

1.2.4 M137A3. This Panoramic Telescope is intended to support the M198 Howitzer and does not use tritium.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-AIS-SS, Picatinny, New Jersey 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not documents are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-13830	-	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	-	Fire Control Materiel, Manufacture, and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

(Copies of federal and military specifications, standards, and handbooks are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

5549108	-	Telescope, Collimating
7680631	-	Dioptometer
8565556	-	Pressure Tester
10549198	-	Adapter, Vibration and/or Shock
10558252	-	Test Fixture
11741101	-	Telescope, Panoramic: M137
11747952	-	Adapter, Torque
11747953	-	Adapter, Torque

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11747954	-	Adapter, Torque
11747955	-	Adapter, Torque
11747975	-	Gage, Interchangeability, Maximum
12599167	-	Telescope, Panoramic: M137A1 (Deflection method of lay)
12599209	-	Telescope, Panoramic: M137A1 (Bearing method of lay)
12984713	-	Telescope, Panoramic: M137A2
12984775	-	Telescope, Panoramic: M137A3

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 9000 - Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to ANSI New York City Office Operations, 25 West 43rd Street, 4 floor, New York, NY 10036 or go online to http://www.ansi.org)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Deleted.

3.2 First article. When specified (see 6.3), a sample shall be subjected to first article inspection and approval (see 4.2).

3.3 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.4 Fabrication. The telescope shall be manufactured in accordance with Drawing 11741101, 12599167, 12599209, 12984713 or 12984775 as applicable and drawings pertaining thereto.

3.5 General specifications. The contractor shall be responsible for the compliance with the requirements of specifications MIL-PRF-13830 and MIL-F-13926.

3.6 Environmental.

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3.6.1 Storage temperature. The telescope shall show no evidence of physical failure when thermally stabilized at ambient temperatures of +160 °F (+71 °C) ± 5°F (±3 °C) and -60 °F (-51 °C) ± 5 °F (± 3 °C) for a period of 4 hours ± 15 minutes at each temperature.

3.6.2 Operating temperature. The telescope shall meet the applicable requirements of 3.5.8 through 3.5.9.2 inclusive while exposed and thermally stabilized at ambient temperatures of +145°F (+63°C) ± 5°F (±3°C) and -50 °F (-46°C) ± 5°F (± 3°C) for a period of 4 hours ± 15 minutes at each temperature. Upon return to standard ambient temperature, +60 °F to +90°F (+16°C to + 32°C), from each operating temperature, the telescope shall meet the requirements of this specification.

3.6.3 Shock. The telescope shall withstand a total of 75 shock impulses, 15 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half sine wave with a time duration and peak amplitude for each shock impulse as follows:

Longitudinal: -50g, +50g both half sine wave .010 sec. +/- .001 sec. duration, 15 each

Vertical: +50g half sine wave .010 sec. +/- .001 sec. duration, 15 each

Transverse: -50g, +50g both half sine wave .010 sec. +/- .001 sec. duration, 15 each

Subsequent to shock, the telescope shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.14.

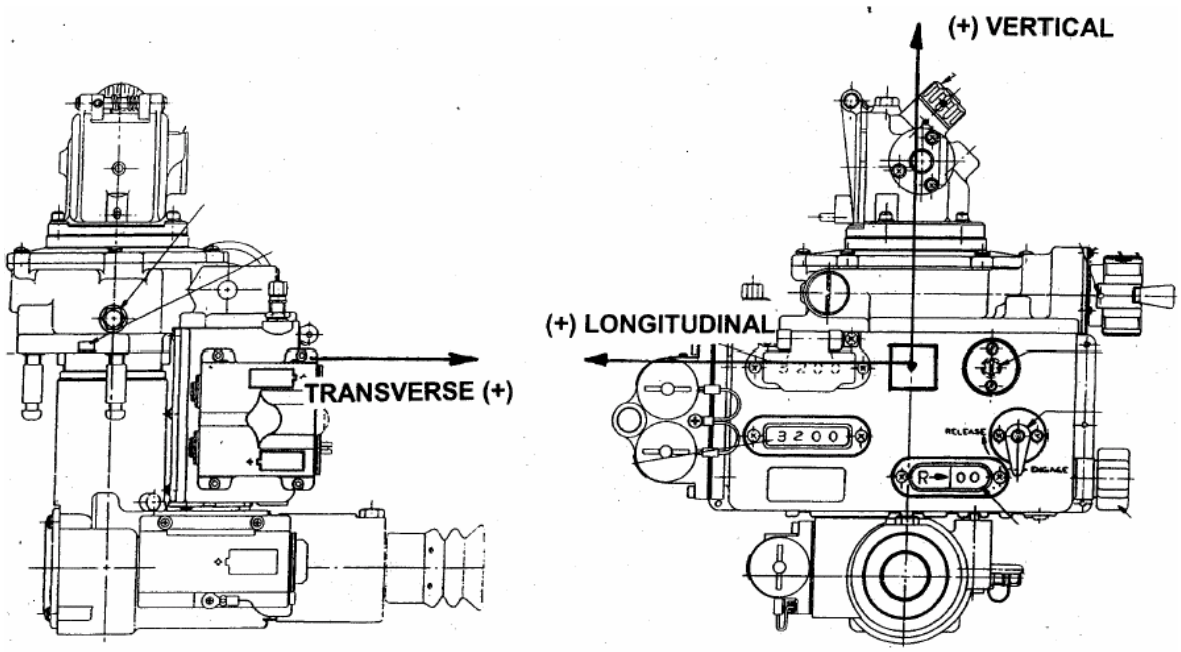


Figure A. Direction of shock loads

3.6.4 Vibration "A". Where required, the telescope shall withstand a total of 270 minutes ± 5 minutes of sweep-cycle vibration. The vibration shall be applied for 90 minutes ± 2 minutes along each of the three mutually perpendicular major axes. A

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complete sweep-cycle shall consist of vibration from origin, 5 hz at 1 inch double amplitude to mid-point 5 g's ± 0.5 g's at 500 hz to origin, and shall have a duration of 15 minutes ± 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz, and varied with frequency to maintain a constant 5 g's ± 0.5 g's acceleration between 10 hz and 500 hz. Upon completion of vibration, the telescope shall exhibit no evidence of damage and shall meet the requirements of 3.7 through 3.14.

3.6.4.2 Vibration "B". Subsequent to meeting the requirements of 3.7 the telescope shall be vibrated in a vertical plane for a total of 15 minutes +/- 1 minute of sweep cycle vibration. A complete sweep cycle shall consist of vibration from origin (5 hz at 1 inch double amplitude) to mid-point (5 g's +/- 0.5 g's at 500 hz) to origin and shall have a duration of 15 minutes +/- 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz and varied with frequency to maintain a constant 5 g's +/- 0.5 g's acceleration between 10 hz and 500 hz. Upon completion of vibration, the telescope shall exhibit no evidence of damage or physical failure and shall meet the requirements of this specification.

3.7 Orientation. The requirements of 3.6 through 3.13.3 inclusive, shall be met with the flat mounting surface of the telescope horizontal, and the eyepiece elbow shall be positioned perpendicular to the plane of the telescope locating keyways. The Azimuth "Upper 6400 mil" and "Lower 6400 mil" counters shall indicate 3200 mils, and lower right correction counter indicating zero. With the head cover open, the line of sight shall be horizontal and parallel to the plane of the locating keyways. The line of sight shall be established by placing the telescope reticle into coincidence with the collimator target reticle. Set the 5 mil click lead knob for indirect fire out of the detent.

3.7.1 Collimation change, post vibration. This requirement shall be met with conditions of 3.7 re-established without readjusting boresight subsequent to vibration. The line of sight established prior to vibration shall not deviate more than .25 mil in deflection due to subjecting the telescope to Vibration "A" or Vibration "B".

3.8 Sealing and purging.

3.8.1 Optical assembly area. The telescope optical assembly area shall be capable of sustaining an internal pressure of 7 + 0.10 pound per square in gage (psig) for a minimum of one hour with no evidence of leakage.

3.8.2 Counterbox area. The counterbox area of the telescope shall be purged with the dry nitrogen supplied at a pressure of 5 + 0.10 psig for a minimum of one hour with no evidence of leakage.

3.8.3 Purging. The optical assembly and couterbox area of the telescope shall be purged with dry nitrogen supplied at a pressure of 8 + ¼ psig until the dew point of the emergent gas is no higher than -25 °F. Purging apparatus shall be removed, when relief valve is reseated and maintains a minimum pressure of 5 psig.

3.9 Cleanliness.

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3.9.1 Dirt. Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle surface shall not exceed the width of the reticle line. The maximum number of dirt particle appearing within a central field of view of 50 mils diameter shall be three (3). The maximum number of dirt particles appearing outside this field of view shall be three (3). The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than ¼ of the reticle line width shall be ignored regardless of distribution.

3.9.2 Other defects. There shall be no evidence moisture, grease, fingerprints, condensates, fractures and adhesive separations when viewing through the objective and eye end of the instrument.

3.10 Optical characteristics.

3.10.1 Eyepiece focus. The best eyepiece focus of the reticle image shall be fixed between the limits of -0.50 and -1.00 diopters. The variation in eyepiece focus between the horizontal and vertical image shall not exceed 0.25 diopter.

3.10.2 Parallax. Parallax at the center of the field shall not exceed 0.15 mil when viewing a target at 80 + 10 meters.

3.10.3 Reticle Accuracy. The angular separation between the 40 mil graduations on the horizontal axis shall be 80 mils ± 0.5 mil.

3.10.4 Resolution. The resolution on the optical axis shall be 15 seconds of arc or less when using an observation telescope of at least 3 power and entrance pupil of at least 0.4 inches. Resolution consists of detectable line structure and proper line count in all four meridians. Total spread in focus shift between meridians shall not exceed 0.25 diopter as measured at the eyepiece.

3.10.5 Parallelism of reticle and image. The vertical reticle line shall be parallel to the image of a vertical target line and shall not exceed 30 minutes of arc at the position establish in paragraph 3.7.

3.10.6 Image tilt. The image of a vertical target line shall be vertical and shall not exceed one degree of arc throughout two (2) revolutions of the head. This requirement shall also be met with the elbow assembly positioned against each of the swing stops.

3.10.7 Reticle cut-off. All reticle graduations, including the “80” numerals shall be visible throughout two (2) complete revolutions of the Head. This requirement shall be met with the eyepiece elbow assembly positioned against each of the swing stops.

3.11 Performance.

3.11.1 Elevation mechanism.

3.11.1.1 Excursion range. Rotation of the elevation knob shall cause the line of

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sight to elevate and depress a minimum of 300 mils from the position specified in 3.7.

3.11.1.2 Plumb travel. Starting with the line of sight in coincidence with a vertical target line at zero elevation, the line of sight shall track the target line within 0.5 mil when elevated 300 mils, depressed to 300 mils depression, then elevated to zero elevation.

3.11.2 Azimuth mechanism.

3.11.2.1 800 mil steps. Error at deflection or bearing readings in 800 mil steps throughout two full revolutions of the head in one direction shall not exceed 1.0 mil including backlash. The total spread between the greatest positive error and the greatest negative error shall not exceed 1.0 mil, including backlash. The total error, excluding backlash, between the 6400 mil upper azimuth counter reading and the deflection or bearing (lower 6400 mil) counter reading for any one number shall not exceed 0.25 mil.

3.11.2.2 15 mil steps. Rotating the correction knob to indicate 15 mils, followed by rotation of the azimuth knob in the opposite direction with backlash removed, to deflect the line of sight 15 mils, shall cause the azimuth counter to return to within 0.25 mil of the original setting. This requirement shall be met with the 15 mil valve set into each “R” and “L” correction counter at azimuth positions 3200, 4800, zero, and 1600 mils as read on the azimuth counter.

3.11.2.3 Level travel. Starting with the line of sight in coincidence with a horizontal target, the line of sight shall track the target within 1.0 mil (total spread of 2 mils) when the head is rotated through two complete revolutions in azimuth.

3.11.2.4 Lift. Vertical displacement of the line of sight due to reversing rotation by means of the azimuth knob shall not exceed 0.5 mil.

3.11.2.5 Five mil click lead mechanism. With the lead knob set to “Direct” fire, a detent action shall be felt and heard when the azimuth knob is rotated to introduce increments of 5.0 mils + 0.5 mil into the 6400 mil azimuth counter. This requirement shall be met at standard and extreme operating temperatures.

3.11.3 Backlash.

3.11.3.1 Azimuth mechanism. Backlash in the azimuth mechanism shall not exceed .5 mil as read on the azimuth counter and .75 mil when read on the deflection or bearing counter when checked at 3200 and 0 mils.

3.11.3.2 Elevation mechanism. Backlash in the elevation mechanism shall not exceed 1.0 mil.

3.11.4 Counter mechanism.

3.11.4.1 Boresight retention. An original boresight setting shall remain within 0.25 mil, as read on the azimuth counter, subsequent to rapid traverse specified in 3.11.4.2.

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3.11.4.2 Rapid traverse. With the correction counter indicating zero, the deflection or bearing counter shall follow the azimuth counter within 0.25 mil with the azimuth knob is rotated at 140 ± 10 rpm for 30 seconds ± 5 seconds in each direction, excluding backlash. The correction counter shall remain zero.

3.11.4.3 Deflection or bearing counter. The counter shall be settable to any reading from 0 to 6399 within + 0.25 mil.

3.11.4.4 Correction counter, excursion range. The correction counters shall have a minimum of 95 mils excursion, but not over 99 mils excursion, on the “L” and “R” dials.

3.11.4.5 Correction counter setting affect on deflection or bearing counter. Setting 95 mils on the “L” counter shall increase the deflection or bearing counter reading 95 ± 0.5 mils. Setting 95 mils on the “R” counter shall decrease the deflection or bearing counter reading 95 ± 0.5 mils.

3.11.4.6 Correction counter setting. There shall be no change greater than .25 mil in the azimuth counter readings and 0.1 mil maximum movement of the line of sight when the 95 mil values are set into the correction counters.

3.11.5 Open sight. With the telescope positioned as in 3.7, the line of sight of the open sight shall be parallel in azimuth and elevation to the line of sight through the telescope optics within 10 mils.

3.11.6 Torque. The running torque required to operate the following knobs shall be within the values specified for standard ambient temperature of +60 °F to +90°F, and shall not exceed the values specified for the extreme operating temperatures. The torque values for the azimuth knob shall be met with the five mil click lead knob set for “Indirect” fire (out of detent).

KNOB	At + 60 ° F to +90 °F	At -50 ° F and +150 °F
Azimuth	3 to 8 in. lbs.	15 in. lbs. max.
Correction counter	10 to 20 in. ozs.	40 in. ozs. max.
Boresight Adjustment	7 to 80 in. ozs.	10 in. lbs. max.
Elevation	5 to 40 in. ozs.	80 in. ozs. max.
Clutch Release	7 to 80 in. ozs.	10 in. lbs. max.

3.11.7 Direct force. The following operability requirements shall be met with the force applied horizontally and in the direction of movement.

3.11.7.1 Elbow assembly. At a standard ambient temperature of +60° to +90°F, a direct force of 1 to 10 pounds, applied adjacent to the eye lens, shall be required to rotate the elbow assembly in each direction through the full extent of travel. The force required at the extreme operating temperatures of 3.6.2 shall not exceed 15 pounds.

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3.11.7.2 Operability. The azimuth counter cover and latch, head cover and latch, azimuth knob folding handle, elbow detent lock pin, and the 5 mil click lead shall be operable at the extreme operating temperatures of 3.6.2.

3.12 Counter numerals. With the units digit of each counter in coincidence with its respective index line, the remaining numbers shall be in horizontal alignment within one-sixteenth of an inch.

3.13 Illumination.

3.13.1 Reticle. The reticle markings shall be clearly distinguishable when observed in ambient light conditions, ranging from dusk into night.

3.13.2 Counters. The numerals on the azimuth, deflection or bearing and correction counters, shall be distinguishable when observed under any ambient light conditions, ranging from dusk to darkness.

3.13.3 External tritium contamination. Any part of tritium shall not be present on the external surfaces of the M137 and M137A1 Panoramic Telescope in excess of 1000 dpm/100 sq.cm. subsequent to requirements 3.6.1 through 3.6.5, as required by final acceptance inspection, 4.3.2.2 and special sampling, 4.4, as applicable.

3.14 Interchangeability. Telescope interchangeability shall be verified by the insertion, seating and subsequent removal of the maximum fit gage into keyways of the telescope mounting surface without binding.

3.15 Reliability. The Panoramic Telescope shall be subjected to reliability assurance tests as follows:

a. The telescope meet the storage and operating requirements of 3.6.1 and 3.6.2.
b. The telescope shall be shocked and conform to the requirements of paragraph 3.6.3
c. The telescope shall withstand the vibration “A” requirements as noted in paragraph 3.6.4.

3.16 Workmanship. The workmanship of MIL-PRF-13830 and MIL-F-13926 shall apply.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed

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requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-PRF-13830 and MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in ISO 8402.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval of tests shall consist of three (3) telescopes plus three (3) each of all items covered by Quality Assurance Provisions (QAPs). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The 3 telescopes shall be tested in accordance with, and meet the requirements of, Tables I and II and paragraph 4.9.2.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, with the sample, including certificates of conformance for materials.

4.3.3 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the government for apporval. The Government reserves the right to witness the contractor's inspection.

4.3 Examination and tests.

4.3.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in QAPs listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed. The tabulated classification of defects in Table I and II shall constitute the minimum inspection to be performed by the

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supplier after first article approval and prior to government acceptance or rejection by item or lot.

4.3.3 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

4.3.4 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provision of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

TABLE I: Requirements and test procedures.

No.	Characteristic	Requirement	Test procedure
Critical: None			
Major: 100% Inspection			
101.	Sealing (optical assy)	3.8.1	4.11.1
102.	Sealing (counterbox assy)	3.8.2	4.11.2
103.	Purging (optical assy)	3.8.3	4.11.3
104.	Purging (counterbox assy)	3.8.3	4.11.3
105	Interchangeability	3.14	4.10
106.	Illumination (reticle)	3.13.1	4.8.22
107	Illumination (counters)	3.13.2	4.8.23
108	External Tritium contamination	3.13.3	4.8.24
109.	Orientation & collimation	3.7	4.7.1. & 4.7.2
110.	Vibration "B"	3.6.5	4.9.4
111.	Collimation Change	3.7.1	4.7.2
112.	Defects (Optical)	3.9.2	4.8.25
113.	Eyepiece focus	3.10.1	4.7.4
114.	Parallax	3.10.2	4.7.5
115.	Reticle accuracy	3.10.3	4.7.6
116	Resolution	3.10.4	4.7.7

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117	Parallelism of Reticle & Image	3.10.5	4.7.8
118.	Image Tilt	3.10.6	4.7.9
119.	Reticle cut-off	3.10.7	4.7.10
120.	Excursion range (elevation)	3.11.1.1	4.8.1
121.	Plumb travel	3.11.1.2	4.8.2
122.	800 MIL steps	3.11.2.1	4.8.3
123.	15 MIL steps	3.11.2.2	4.8.4
124.	Level travel	3.11.2.3	4.8.5
125.	Lift	3.11.2.4	4.8.6
126.	5 MIL click lead	3.11.2.5	4.8.7
127.	Backlash (azimuth)	3.11.3.1	4.8.8
128.	Backlash (elevation)	3.11.3.2	4.8.9
129.	Boresight retention (counter)	3.11.4.1	4.8.10
130.	Rapid Traverse	3.11.4.2	4.8.11
131.	Deflection or bearing counter	3.11.4.3	4.8.12
132.	Correction counter (excursion)	3.11.4.4	4.8.13
133	Correction counter (affect on Deflection or bearing counter)	3.11.4.5	4.8.14
134.	Correction counter setting	3.11.4.6	4.8.15
135.	Open sight	3.11.5	4.8.16
136.	Torque	3.11.6	4.8.17
137.	Elbow assembly (direct force)	3.11.7.1	4.8.19
138.	Operability	3.11.7.2	4.8.20
139.	Counter numerals	3.12	4.8.21
140.	Dirt	3.9.1	4.8.25
141.	Fabrication	3.4	4.1.1
142.	Workmanship	3.16	MIL-F-13926
143.	Packaging	5.1	4.12

Minor: None

4.4 Special sampling.

4.4.1 General. Subsequent to meeting the requirements of Table I, three mounts shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table I, except characteristic 110, after being subjected to Table II testing.

TABLE II. Classification of Defects.

No.	Characteristic	Requirements	Test Procedure
301.	Storage temperature	3.6.1	4.9.1
302.	Operating temperature	3.6.2	4.9.1
303.	Torque	3.11.6	4.8.17
304.	Elbow assembly (direct force)	3.11.7.1	4.8.19

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4.4.2 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary correction have been made by the contractor and the resubmitted samples have been approved.

4.5 Test equipment.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.3 Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in Specifications or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the TDP.

4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is considered as part of the prescribed product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribe product tolerance limit. Thus, the design of the test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3.3 Test equipment. In conjunction with 4.5.3.2, the following standard test equipment shall be utilized in the performance of the applicable test as specified in 4.6.

Nomenclature	Description
1. Hot and cold chamber	Standard type conforming to the accuracies specified under Test Facilities of MIL-F-13926.
2. Light Source Standard	
3. Torque wrench	Standard shop type accurate to 10 % of the drawing tolerance.

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4.6 Reliability assurance sample. Unless otherwise specified in the contract, the Government shall be responsible for reliability assurance testing. The reliability assurance test sample shall consist of one telescope selected at random by the Government from any accepted regular production lot prior to delivery of the first 25% of the basic contract quantity. All telescopes previously selected for First Article Test or Special Sampling shall be excluded from the lot for purposes of selecting the reliability assurance test sample.

4.6.1 Reliability assurance testing. After random selection of the sample, the contractor shall provide to the Government the inspection records, including certificates of conformance, for the sample. The contractor shall also provide a list of all changes, deviations, and waivers under the contract. The sample shall be tested for reliability as represented by the requirements of 3.15. Testing shall be performed in accordance with the procedures specified in 4.9.1 through 4.9.3.

4.6.2 Defects. Defects resulting from reliability assurance testing may be assessed as sufficient evidence that the contractor's production processes and quality control procedures do not provide adequate assurance against product reliability degradation. Upon identification by the government to the Contractor, of the specific quality defect(s) found, the contractor shall determine and apply effective corrective action to improve his production processes and quality control procedures as necessary to eliminate the defect(s) from telescopes not yet delivered under the contract.

4.6.3 Retest. Whenever reliability assurance test results require corrective action by the contractor, a retest sample shall be selected in accordance with 4.6 except that the sample shall be selected from the first 10 telescopes produced subsequent to implementation of the corrective action. The retest sample shall be subjected only to the reliability test(s) which revealed the defect(s).

4.6.3.1 Inspection of the sample after retest shall be conducted by the Government and shall be in accordance with 4.3.2, Final Acceptance Inspection, except characteristics 110 of Table I, and characteristics 301 and 302 of Table II. Reappearance of the previous defect(s), 4.6.2, may be cause for the Government to prescribe mandatory process corrective action by the Contractor at no additional cost to the Government.

4.7 Methods of inspection.

4.7.1 Orientation. Test fixture 10558252 shall be used for the inspection of the Panoramic Telescope M137, M137A1, M137A2 and M137A3. Position the test fixture on a vibration free surface in accordance with the set-up instructions on Drawing F10558252. Follow the calibration and set-up instruction prior to positioning the telescope to the fixture. Subsequent to completion of all set-up instructions, the telescope shall be oriented as specified in paragraph 3.7 of this specification.

4.7.2 Collimation. The telescope shall be positioned on fixture 10558252 with the flat mounting surface horizontal, the eyepiece elbow shall be positioned perpendicular to the plane of the telescope locating keyways. The azimuth counter shall be set at 3200 mils

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and the deflection or bearing counter shall be set at 3200 mils. The correction counter shall indicate zero. The head cover shall be in the open position with the line of sight being to the left of the elbow eyepiece. The 5 mil click lead actuator (located in azimuth knob) shall be set for indirect fire (out of detent). The telescope reticle shall be placed into coincidence with the collimator target reticle 7573291. The line of sight of the telescope shall be in the exact coincidence with the target reticle, the position recorded, prior to all vibration tests.

4.7.3 Collimation post vibration. Subsequent to vibration testing, the line of sight shall not have deviated more than the deflection allowed in 3.7.1.

4.7.4 Eyepiece focus. The telescope eyepiece is a fixed focus assembly and it shall be set between the tolerances specified in 3.10.1. To determine the correct setting of the eyepiece, dioptometer 7680631 of three power (3X) shall be used that has been adjusted to the individual eye. This can be accomplished by sighting trough the dioptometer at a white background and adjusting the eyepiece of the dioptometer until the reticle is at the sharpest point obtainable. Position the barrel of the dioptometer to set the index at zero. When this has been accomplished, sight through the dioptometer and the eyepiece of the telescope. Observe the reticle of the telescope, and compare it with the reticle of the dioptometer. Adjust the barrel of the dioptometer to make the reticle of the telescope go out of focus; then re-adjust the barrel to bring the telescope reticle into focus at the sharpest point. Read the error on the scale on the barrel of the dioptometer.

4.7.5 Parallax. With the telescope mounted on the 10558252 test fixture, and the collimator projector set to a range of 80 ± 10 meters, turn the azimuth knob until telescope range is superimposed on the target reticle. Observe any apparent movement between the telescope and target reticle as the observers head is moved up and down and from side to side within the limits of view imposed by the telescope eyeshield. The apparent motion shall not exceed the limits of paragraph 3.10.2.

4.7.6 Reticle accuracy. With the telescope oriented as described in paragraph 4.7 adjust the fixture rotary table so that the right 40 mil mark of the telescope is superimposed on the target centerline. Read the azimuth counter. Rotate the azimuth knob until the left 40 mil mark is superimposed on the target centerline. Reread the azimuth counter. The difference in counter readings is the angular separation between 40 mil graduations and shall be within the limits of paragraph 3.10.3.

4.7.7 Resolution. Place viewing telescope 7680631 in front of the panoramic telescope's eyepiece. The wall target and testing procedures for resolution shall be in accordance with MIL-PRF-13830. The telescope shall not exceed the limits specified in 3.10.4.

4.7.8 Parallelism of the reticle and image. When oriented per paragraph 4.7, the azimuth knob shall be used to position the vertical reticle line of the telescope so that it passes through the intersection of the vertical target line and the short horizontal mark on the target reticle (about 20 mils below its center cross). When so positioned, the telescope vertical reticle line shall lie within the 30 minute limit lines of the target reticle to display conformance with paragraph 3.10.5. This test shall also be repeated with the elbow

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positioned against the right and left swing stops and at azimuth settings of 3200 mils, 4800 mils, 6400 mils and 1600 mils.

4.7.9 Image tilt. The telescope shall be orientated in accordance with 4.7. Use viewing telescope 5549108 and a flat surface to position the viewing telescope at the eyepiece of the telescope under test, adjust the viewing telescope to the height of the telescope eyepiece. Adjust the viewing telescope's position so that the vertical reticle line is in coincidence with the target vertical reticle line at the short horizontal mark (about 20 mils below the center cross). The vertical target line shall be parallel to the vertical line of the viewing telescope within the limits of the 1 degree mark of the target telescope per 3.10.6. This test shall be repeated with the elbow positioned against the right and left swing stops and at the azimuth settings of 3200 mils, 4800 mils, 6400 mils and 1600 mils.

4.7.10. Reticle cut-off. This test shall be performed by viewing through the eyepiece of the telescope while turning the azimuth knob two complete revolutions of the head. The reticle graduations and numerals shall be as specified in 3.10.7.

4.8 Performance.

4.8.1 Excursion range (elevation mechanism). With the line of sight established in accordance with 3.7, elevation and depression of the prism in the rotating head by means of the elevation knob shall cause the line of sight to displace the amount specified in 3.11.1.1 for elevation and depression. This test can be accomplished by using a wall target (see sketch Figure 1) that has an established level line of sight and the excursion limits required.

4.8.2 Plumb travel. This test can be performed concurrently with 4.8.1. The telescope shall be in a level position with the vertical reticle line brought into coincidence with the vertical target line by rotating the azimuth knob. The horizontal reticle of the telescope shall be in coincidence with the horizontal target line by rotating the elevation knob. Rotate the elevation knob to elevate the line of sight of the telescope 300 mils. Reverse the rotation of the knob to depress the line of sight to 300 mils depression. Reverse the rotation again, returning the line of sight to zero elevation. The vertical reticle line of the telescope must track the target line within the tolerance specified in 3.11.1.2.

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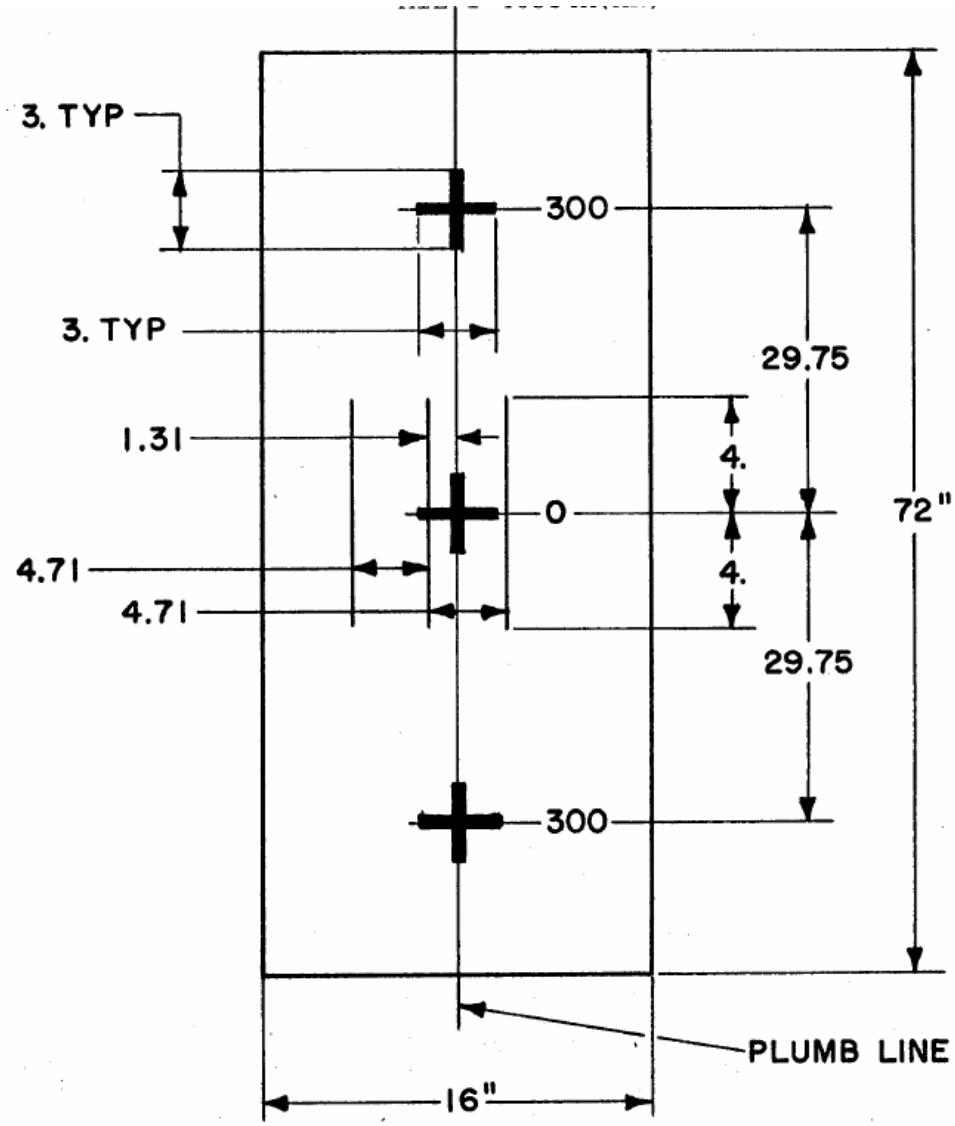


FIGURE I. Excursion range and plumb travel target

4.8.3 800 mils steps. Position the telescope as specified in 3.7. Set the telescope azimuth counters to 3200 Mils and the correction counter to zero. The vertical line on the reticle of the telescope shall be coincident with the vertical line on the target, and the horizontal line of the reticle shall be coincident with the zero elevation line. Azimuth readings shall be made at 800 mil increments through two full revolutions in one direction. The error in the readings taken shall not exceed the tolerance specified in 3.11.2.1.

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4.8.4 15 mil steps. Position the telescope as specified in 4.8.3. Subsequent to orientation of the line of sight and setting the counters, the deflection or bearing counter Direct Fire Knob shall be in the “Engage” position. The 15 mil step test shall meet the requirement of 3.11.2.2 at the “L” and “R” directions and also at 3200, 4800, zero, and 1600 mil azimuth position.

4.8.5 Level travel. This test shall be accomplished with the test procedure of 4.8.3. During the azimuth reading for each 800 mil increment, the position of the horizontal reticle line of the telescope shall be observed with respect to the zero elevation line on the target. The telescope reticle shall track the zero elevation line on the target within the tolerance specified in 3.11.2.3 for at least two revolutions in azimuth

4.8.6 Lift. The vertical and horizontal lines on the telescope reticle shall be positioned on the collimator target as in the initial position of 3.7. The azimuth knob shall be rotated, first in one direction and then in the reverse direction. Lift is indicated if the line of sight is vertically displaced from the original zero elevation setting, and it shall not exceed the tolerance specified in 3.11.2.4. This examination shall be repeated at the 3200, 4800, 6400, and 1600 mil settings.

4.8.7 Five mil click mechanism. Set the lead actuator knob in the Direct fire position and rotate the azimuth knob through the full range of operation. During rotation, a tactile examination shall be made to feel the detent action required in 5 mil increments. An audible examination shall be simultaneously performed to listen for a distinct click resulting from positive detent seating. At least five readings throughout the range of operation, the displacement from one increment setting to the next shall represent a 5 mil angular change within the tolerance specified in 3.11.2.5. All the above examinations shall be performed at standard ambient temperature (+60° +90°F) and the extreme operating temperatures of 3.6.2 to determine compliance with 3.11.2.5.

4.8.8 Backlash (azimuth). The azimuth knob shall first be rotated in one direction to bring the vertical line of the telescope reticle in coincidence with the azimuth zero point on the target. Record the reading of the azimuth counter and deflection or bearing counter. Continue the rotation of the knob in the same direction, one complete turn, then reverse the direction of the knob while sighting through the telescope, reposition the vertical reticle line of the telescope in coincidence with the azimuth zero point on the target, without overtravel. Take readings from the azimuth counter and the deflection or bearing counter then compare them with the original set of readings for any difference in the amount of backlash in the azimuth mechanism. The backlash shall not exceed the tolerances specified in 3.11.3.1 when checked at 3200 and zero mils.

4.8.9 Backlash (elevation). The test for backlash in the elevation mechanism shall be accomplished concurrently with the tests indicated in 4.8.1 and 4.8.2. The horizontal reticle line of the telescope shall be brought into coincidence with the horizontal target line, at zero elevation without overtravel, when the elevation knob is rotated in a clockwise direction. The zero line on the knob and the index line on the rotating head shall be set alignment with each other. Rotate the elevation knob clockwise to 300 mils, then reverse direction of the knob and align the zero line on the knob with the index line on the head without overtravel. Observe, through the telescope, the position of the telescope reticle on

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the target. The telescope reticle shall be within the tolerance specified in 3.11.3.2 of the zero level line on the target. This test shall also be conducted in the counterclockwise direction.

4.8.10 Boresight retention. Position the telescope as specified in 3.7, the azimuth and deflection or bearing counters shall be set at 3200 mils. Rotate the azimuth knob, remove backlash and establish a line of sight in coincidence to the target collimator. Turn the azimuth knob at the speed and for the length of time specified in 3.11.4.2 until the head has made one revolution. Caution should be taken to avoid overtravel in the line of sight coincidence previously obtained. The boresight setting shall read 3200 mils with the specified limit of 3.11.4.1 as read on the azimuth counter. Repeat the above test in the opposite direction.

4.8.11 Rapid traverse. This test may be performed in conjunction with the boresight retention test 4.8.10. Set the correction counter to zero. Turn the azimuth knob at the speed and rotate one complete revolution. The deflection or bearing counter shall follow the azimuth counter within the limit specified in 3.11.4.2. Backlash shall be removed prior to traversing the azimuth knob for this test and the boresight retention test. Rapid traverse shall be repeated in the opposite direction.

4.8.12 Deflection or bearing counter. Set the azimuth and deflection or bearing counters to read zero. Place the deflection counter lever in the “Engage” position. Turn the azimuth knob to 800, 1600, 2400, and 3200 mil settings to determine if above azimuth settings can be set to the accuracy specified in 3.11.4.3.

4.8.13 Correction counter excursion. The correction counter knob shall be manually rotated until full excursion has traveled the minimum and maximum limits specified in 3.11.4.4.

4.8.14 Correction counter affect on deflection or bearing counter. Set the deflection counter and the correction counter in the zero position. Turn the correction counter knob either clockwise or counterclockwise direction. Knob rotation shall increase or decrease the deflection or bearing counter readings in accordance with the values specified in 3.11.4.5.

4.8.15 Correction counter setting. The panoramic telescope shall be positioned as specified in 3.7 and 3.7.1. While sighting through the telescope eyepiece, turn the correction knob to 95 mils, “L” direction. There shall be no change in the line of sight greater than specified in 3.11.4.6. The azimuth counter shall not changed more than the specified limit in 3.11.4.6. This test shall be repeated in the “R” direction.

4.8.16 Open sight. The line of sight shall be set at zero elevation. Wall target (see Figure 1) shall be positioned approximately 8 feet from the objective head window. The wall target shall be plumbed by a plumb line placed in front of the wall target. The line of sight through the open sight shall be parallel in deflection to the optical line of sight within the tolerance limits specified on the wall target and 3.11.5.

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4.8.17 Torque. The running torque required to rotate the knobs and level specified in 3.11.6, at standard ambient and extreme operating temperatures, shall be determined with a standard torque measuring device. The torque measuring device shall be equipped with the following adapters to fit the respective knobs and lever being tested. The measured torque values shall be within the limits specified in 3.11.6.

KNOB	ADAPTER NUMBER
Azimuth	11747952
Correction Counter	11747953
Boresight Adjustment	Screw Driver Tip (std)
Elevation	11747954
Clutch Release (lever)	11747955

4.8.18 Direct force.

4.8.19 Elbow assembly. A standard force measuring device shall be used for this test. The force required to rotate the elbow assembly in both directions shall not exceed the values specified in 3.11.7 and 3.11.7.1.

4.8.20 Operability. The azimuth counter cover and latch, head cover and latch, azimuth knob folding handle, the elbow detent lock pin, and the 5 mil clock lead shall be operable at room temperature and the extreme operating temperatures specified in 3.6.2 and 3.11.7.2.

4.8.21 Counter numerals. A standard measuring scale shall be used to determine numeral (digit) alignment to the respective index line within the linear value specified in 3.12.

4.8.22 Reticle illumination. The reticle illumination test shall be performed under simulated conditions specified in 3.13.1. Place the instrument in a dark room having a controlled light source. The observer and instrument must become dark adapted. Prior to performing the test, place a white matted surface approximately 2 feet from the telescope window. Perform the test by observing reticle pattern and determine that this pattern is clearly discernible by means of the self-illuminating source in the dark room. Gradually raise the level of illumination to where the pattern becomes indiscernible. Place a fixed stop having an aperture of one-half inch over the objective window and observe that the reticle pattern in again clearly discernible by means of the self-illumination to 10 foot candles, remove the fixed stop, and observe the reticle pattern.

4.8.23 Counter illumination. Follow 4.8.22 to acquire proper dark adaptation. The unaided eye shall be capable of determining the counter numerals and numeral alignment at various light levels that simulate dusk into darkness.

4.8.24 Radiological contamination. Contamination test for the M137 and M137A1 shall be performed by wiping the telescope with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper

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shall be placed in a container with the proper portion of scintillation liquid. The scintillating system used to measure contamination must be calibrated and be within 10% of the known standard value when counted to a total of 2000 disintegrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.13.3.

4.8.25 Cleanliness. The unit under test shall be subjected to the test procedures outlined in MIL-PRF-13830 for conformance to requirements outlined in 3.9.1.

4.9 Environmental conditions.

4.9.1 Storage and operating temperatures. The telescope shall be subjected to one cycle of temperatures variations in accordance with Table III. Upon completion of sequence 1 and 4 soak periods (Storage), the telescope shall be visually and tactually examines to insure conformance with the requirements of 3.7.1. Upon completion of Sequence 2 and 4 soak periods (operating), the telescope shall be visually and tactually examined to assure conformance with requirements of 3.12. Following completion of Sequence 3 and 5 soaks periods (Room temperature) the telescope shall meet all the requirements of 3.7.6 through 3.13.1.

TABLE III. Temperature Cycle

	STORAGE		OPERATING			
SEQ	TEMP	SOAK	TEMP	SOAK	ROOM TEMP	SOAK
1	+160°F (+71°C)	4 hrs				
2			+150°F (+65°C)	2 hrs		
3					+60°F to +90°F (+16°C to 32°C)	3 hrs
4	-50°F (-46°C)	4 hrs	-50°F (-46°C)	4hrs *		
5					+60°F to +90°F (+16°C to 32°C)	3 hrs

*Same 4 hours as storage soak

4.9.2 Shock. This test is applicable to the First Article Samples and the Reliability Samples only. The telescope shall be positioned on Adapter 10549198 and secured to the shock tester. The telescope shall be subjected to a shock test in accordance with 3.6.3. Upon completion of the shock test, the mount shall be inspected to the requirements specified in 3.6.3.

4.9.3 Vibration “A”. This test is applicable to the Reliability Assurance Sample only. The telescope shall be positioned on Adapter 10549198 and secured to the vibration tester. The telescope shall be subjected to a vibration test in accordance with 3.6.4 of this specification. Upon completion of the vibration test, the telescope shall be examined and inspected to the requirements of 3.6.4.

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4.9.4 Vibration "B". The telescope shall be positioned on adapter 10549198 and vibrated in accordance with 3.6.5. At the conclusion of the test, the telescope shall be subjected to a visual and tactile examination and shall meet the requirements of 3.6.5.

4.10 Interchangeability. The interchangeability gage, 11747975, shall be applied to the mounting surface (keyway) of the panoramic telescope to assure maximum fit conditions. The gage shall be firmly seated in the keyway and the bolts seated against their respective stops. Any telescope which does not accept the interchangeability gage, without using force, shall be rejected.

4.11 Sealing and purging.

4.11.1 Optical assembly. The telescope shall be capable of withstanding an internal pressure of 7 + 0.10 psig for a period of 1 hour with no evidence of leakage. Prior to the start of the test, secure the poppet valve in a closed position. Then pressurize the telescope with dry nitrogen as specified in paragraph 3.8.1. Accuracy of the test shall be equal to or exceed the accuracies depicted on Pressure Test Drawing 8565556.

4.11.2 Counterbox assembly. The telescope's counterbox assembly shall be capable of withstanding an internal pressure of 5 + 0.10 PSIG for a period of one hour with no evidence of leakage. Prior to the start of the test, secure the poppet valve in the closed position. Pressurize the telescope's counterbox assembly with dry nitrogen as specified in 3.8.2. The accuracy of the test equipment used for this pressure test Drawing 8565556.

4.11.3 Purging assembled telescope. Connect the test equipment to the purging valve of the telescope. Adjust the dry nitrogen pressure to the pressure specified in 3.8.2. The pressure is to remain constant until the dew point of the emergent gas is no higher than -25°F. Disconnect the purging apparatus when the relief valve has reseated to maintain an internal pressure of 3.8.3.

4.12 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of 5.1.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with MIL-STD-2073-1 and the Packaging Data Sheet 11741101, 12599167 or 12599209 as applicable. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended Use. The Panoramic Telescope, M137, M137A1, M137A2 and M137A3 when mounted on their respective telescope mounts are primarily intended to lay the howitzer in azimuth for indirect fire.

6.2 Acquisition requirements. Procurement data should specify the following:

a. Title, number and date of this specification.
b. Selection of an applicable level of preservation, packaging, and packing.
c. Applicable packaging data sheet number (See 5.1).
d. Applicable stock number.
e. Provisions for First Article Testing.
f. Reliability Assurance Sample.
g. Contract data requirements for submission of inspection equipment designs conforming to Data Item Description DI-R-1714. (See 6.3.1)
h. Prospective bidders should be aware that a Nuclear Regulatory License is required for the manufacture of the M137 and M137A1.

6.3 Inspection equipment design.

6.3.1 Submission of designs for approval. Contractor designs for final acceptance inspection must be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List DD Form 1433 in the contract. Unless otherwise specified, data item DI-R-1714 will apply. When the contractor submits inspection equipment designs to the government for approval, he must give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc).
c. The designs remaining to be submitted and the expected date of submittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament, Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by there activities is now under the cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control
Panoramic Telescope
Radioluminous

6.6 Hazardous material.

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6.6.1 Tritium. The M137 and M137A1 Telescope, Panoramic contain tritium, a radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, processing, handling and shipping of radioactive material.

6.6.2 Lithium Batteries. The M137A2 and M137A3 utilize Lithium batteries. Prospective Bidders must be cognizant of the need to handle and/or dispose of these batteries in accordance with all Local, State and Federal Laws.

6.7 Packaging data sheet drawings. The following packaging data sheet drawings should be obtained from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIL-P B455, Picatinny, NJ 07806-5000.

11741101	-	Packaging of Telescope, Panoramic: M137
12599167	-	Packaging of Telescope, Panoramic: M137A1 (Deflection method of lay)
12599209	-	Packaging of Telescope, Panoramic: M137A1 (Bearing method of lay)
SPI12984713	-	Packaging of Telescope, Panoramic: M137A2
SPI12984775	-	Packaging of Telescope, Panoramic: M137A3

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army-AR

Preparing activity:
Army-AR
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